

VEDANTH ANIL WARRIER

27a Cook Street, Baulkham Hills, 2153, Australia
0431906777 | vedanth.warrier@gmail.com
linkedin.com/in/vedanth-warrier
github.com/vedanth-warrier

PROFESSIONAL SUMMARY

Second-year Aerospace Engineering and Mathematics student at UNSW (WAM 88.7) with hands-on experience in SolidWorks FEA, fabrication, aerodynamic testing, embedded systems, and Python numerical simulation of flight mechanics and orbital dynamics. Proven ability to execute engineering projects independently and in team settings.

EDUCATION

UNSW Sydney

Bachelor of Advanced Science (Hons) / Bachelor of Engineering (Hons)

Kensington, Sydney

2025 – Present

- **Majors:** Aerospace Engineering & Mathematics
- **Current WAM:** 88.727 (High Distinction Average)
- **Academic Record:** Engineering Mathematics 2E (96), Computing 1A - C Programming (92), Discrete Mathematics (92), Numerical Methods & Statistics (93), Engineering Mechanics (90), Fluid Mechanics for Engineers (89).

Normanhurst Boys High School

Higher School Certificate (HSC) | ATAR: 99.65

Normanhurst, Sydney

2019 – 2024

ENGINEERING PROJECTS

Orbital Trajectory & Telemetry Suite

Solo Developer

Independent

May 2026 – Jun 2026

- **IVP Trajectory Simulator:** Numerically integrated a **5-state ODE system** (position, velocity, mass) in polar coordinates using `scipy.integrate.solve_ivp` (RK45), modelling gravity-turn ascent with **dynamic atmospheric drag** interpolated from a real altitude-density dataset across **1000 km** of scale height.
- **BVP Orbital Insertion Solver:** Formulated a **3-DOF boundary value problem** to solve for circular orbit insertion parameters across a **3-phase mission** (ascent burn, coast, apogee kick), solved using `scipy.optimize.least_squares` with dynamic variable scaling.

Go-Kart Engineering & Fabrication

Designer & Fabricator

Independent

May, 2025 – Present

- **Design & Simulation:** Engineered the primary chassis and battery holding beams using SolidWorks static FEA, validating a 150kg occupant load and a 100kg battery allowance (Load Factor > 2). Confirmed a **Factor of Safety > 2** (peak stress of 120 MPa against a 250 MPa yield strength) and restricted maximum deflection to 0.9cm, preserving a safe 20cm operational ground clearance.
- **Manufacturing & Assembly:** Executed chassis and sheet metal enclosure fabrication using **Gasless MIG Welding**, and soft soldering for the integration and wiring of 6 12V 22Ah COTS batteries.
- **Testing & Optimisation:** Conducted physical testing and powertrain optimisation, diagnosing a drivetrain chain-slip fault via slow-motion video analysis achieving an acceleration of **4.5 m/s²**

Morphing Wing Aerodynamic Prototype

Co-Designer / Systems Integration

UNSW Sydney

Feb 2025 – May 2025

- **Design:** Designed and 3D-printed a scale morphing wing prototype using SolidWorks, integrating Arduino Uno-controlled servo motors to dynamically alter air foil geometry achieving a maximum **wing angular displacement of 43 degrees**. Iteratively refined through laser-cut acrylic failure analysis before finalising a PLA compliant chevron mechanism.
- **Aerodynamic Testing:** Conducted empirical wind-tunnel testing to validate performance metrics, achieving **~13% increase in the L/D** from 5.00 to 5.68.
- **Control:** Developed an **open-loop direct-mapping** control system to actuate servo motors via potentiometer input, enabling real-time wing profile adaptation.

OTHER EXPERIENCE

Project Academy

HSC Mathematics Tutor

Sydney

Dec 2024 – Present

- Lead groups, maintain classroom control, and break down complex mathematical concepts for senior students.

TECHNICAL SKILLS

Engineering Software & Programming

- **CAD & FEA:** SolidWorks (Proficient in static FEA, complex assemblies, and 3D component modelling)
- **Numerical Analysis:** Python (Linear systems, root-finding algorithms, IVPs, BVPs, PDEs, and data visualisation)
- **Embedded Systems:** C/C++ (Arduino control logic, mechatronic actuation, and hardware-software integration)
- **Aerodynamic Analysis:** XFOIL (Viscous/inviscid airfoil analysis, polar generation, and NACA profile optimisation)

Hardware & Fabrication

- Gasless MIG Welding, Soft Soldering, Surface Mount (SMD) Soldering.

COMMUNITY & FITNESS

- Ultimate Frisbee & Badminton Team Captain (2022–2024); Push-Up Challenge and World's Greatest Shave participant.